# Supplementary Information

Separating decision rule failures from representation failures in external validation of diabetic retinopathy grading

## Contents

**Supplementary Table S1. Reporting checklist (TRIPOD+AI and CLAIM)** **2**

**Supplementary Table S2. Configurations** **3**

**Supplementary Table S3. Pre-registered interventions** **3**

**Supplementary Note 1. Decision rules** **4**

**Supplementary Note 2. The DME grade cannot be derived from published lesion masks** **4**

**Supplementary Note 3. A per-grade verdict that depends on the cut-point procedure** **4**

**Supplementary Note 4. Correction of an archived ordinal-head comparison** **5**

**Supplementary Note 5. Seed variation (O1–O3)** **5**

**Supplementary Note 6. Reverse direction (O4–O6)** **6**

**Supplementary Note 7. Content-overlap audit** **7**

**Supplementary Tables S7–S8. Full decision-rule matrices** **8**

## Supplementary Table S1. Reporting checklist (TRIPOD+AI and CLAIM)

The study develops and externally evaluates prediction models (TRIPOD+AI) built from medical images (CLAIM). The table maps each reporting domain to the place where it is addressed. The journal's official checklist forms are to be completed by the authors at submission from this mapping.

| Domain | Where and how it is reported |
|---|---|
| Title and abstract | Title names the task (DR grading) and the finding; abstract states data, models, external sets, decision rules and main results. |
| Background and objectives | Introduction: the gap (external failures attributed to representations without matched decision rules) and the pre-registered questions. |
| Data sources | Methods, *Data*; Table 1. Public corpora: IDRiD, Messidor-2 (adjudicated grades), EyePACS (pretraining only), APTOS 2019, DDR. |
| Participants and eligibility | Methods, *Data*: every gradable image with a DR grade 0–4; images with DDR grade 5 (ungradable) excluded. No patient-level demographics are released by the corpora; this is stated as a limitation. |
| Data preparation | Methods, *Models and training*: retinal-disc crop, cache at long side 560 pixels, resizing, augmentation. |
| Outcome (reference standard) | International 5-grade DR scale; DME risk grade (IDRiD); grading procedure per corpus in Table 1. |
| Sample size | Fixed by the public corpora; uncertainty reported by bootstrap intervals throughout. |
| Missing data | Messidor-2 has no three-grade DME label; the DME loss is masked for those images (Methods). |
| Data partitions | Methods, *Cross-validation and model selection*: frozen 5-fold split of the development pool; external sets never used for training, selection or threshold fitting except the explicitly diagnostic R4/R4m. Content-overlap audit: Supplementary Note 7. |
| Model specification | Methods and Supplementary Table S2 (every configuration, one factor changed per run). |
| Model output and decision rule | Methods, *Decision rules*; Supplementary Note 1. |
| Performance measures | QWK, Cohen's $\kappa$, accuracy, per-grade recall, macro-averaged recall, referable-DR sensitivity and specificity, AUROC, Brier score, ECE (Methods, *Statistics*). |
| Uncertainty | Paired bootstrap over images (2,000 resamples) for every comparison. |
| Class imbalance | DR-grade-balanced sampler (Methods); per-grade recall reported for every grade. |
| Fairness | No subgroup attributes are released by the corpora; between-population behaviour is studied through four external populations in two directions. |
| Pre-registration | Outcomes O1–O9 and the interventions were registered in the public repository before the corresponding runs (Methods, *Pre-registration*); falsified outcomes are reported. |
| Code and data availability | Data availability and Code availability statements. |
| Ethics | Ethics statement (public, de-identified data). |
| Use of AI assistance | Methods, final paragraph. |
| Limitations | Discussion. |

# Supplementary Table S2. Configurations

**Table S2: Configurations.** Each configuration changes one factor relative to its control. Dev QWK: development out-of-fold DR QWK at matched calibration (fold-wise cross-fitted cut-points); folds: development folds trained.

| Model | Control | Backbone | DR head | Pixels | Pretraining | Epochs | Factor | Dev QWK |
|---|---|---|---|---|---|---|---|---|
| E08 | – | DenseNet121 | cumulative | 448 | ImageNet + EyePACS (5 ep) | 40 | reference | 0.8646 |
| E09 | E08 | DenseNet121 | cumulative | 224 | ImageNet + EyePACS | 40 | resolution | 0.8389 |
| E10 | E08 | DenseNet121 | cumulative | 640 (560 effective) | ImageNet + EyePACS | 40 | resolution | 0.8749 |
| E17NAT | E10 | DenseNet121 | cumulative | 640 | ImageNet + EyePACS | 40 | native-resolution source files | 0.8662 |
| E11FULL | E10 | EfficientNet-B3 | cumulative | 448 | ImageNet + EyePACS | 30 | backbone (selected model) | 0.8954 |
| E14MAC | E08 | DenseNet121 | cumulative | 448 + macula crop | ImageNet + EyePACS | 40 | fovea-centred crop for DME head | 0.8595 |
| E15LPFT | E08 | DenseNet121 | cumulative | 448 | ImageNet + EyePACS | 40 (LP 5) | linear probe then fine-tune | 0.8655 |
| E20CORAL | E08 | DenseNet121 | CORAL | 448 | ImageNet + EyePACS | 40 | ordinal head | 0.8569 |
| E22CORN | E08 | DenseNet121 | CORN | 448 | ImageNet + EyePACS | 40 | ordinal head | 0.8564 |
| E21CNXA | E08 | ConvNeXt-tiny | cumulative | 448 | ImageNet + EyePACS | 40 | backbone (2 folds) | 0.8267 (2 f) |
| I24FULL | E09 | RETFound ViT-L/16 | cumulative | 224 | RETFound (colour fundus) | 30 (LP 5) | foundation encoder | 0.7976 |
| S1E08–S4E08 | E08 | DenseNet121 | cumulative | 448 | ImageNet + EyePACS | 40 | seed only | – |
| R1APTOS | E08 | DenseNet121 | cumulative | 448 | ImageNet + EyePACS | 40 | trained on APTOS (reverse) | – |

# Supplementary Table S3. Pre-registered interventions

**Table S3: Pre-registered interventions, shipped and matched calibration.** Difference (intervention minus control) in development DR QWK with 95% paired-bootstrap intervals (DME QWK for the macula crop; APTOS DR QWK for the ensemble).

| Intervention | Control | Folds | Matched difference [95% CI] | Verdict |
|---|---|---|---|---|
| EfficientNet-B3 backbone | E10 | 5 | +0.0207 [+0.0105, +0.0320] | confirmed |
| Macula-centred crop (DME QWK) | E08 | 5 | −0.0237 [−0.0566, +0.0094] | falsified |
| Linear probe then fine-tune | E08 | 5 | +0.0008 [−0.0105, +0.0126] | falsified |
| Native 640-px resolution | E10 | 5 | −0.0086 [−0.0199, +0.0025] | falsified |
| CORAL ordinal head | E08 | 5 | −0.0076 [−0.0184, +0.0032] | falsified |
| CORN ordinal head | E08 | 5 | −0.0082 [−0.0207, +0.0036] | falsified |
| ConvNeXt-tiny backbone | E08 | 2 | −0.0362 [−0.0593, −0.0153] | falsified, significantly worse |
| RETFound ViT-L/16 (fine-tuned) | E09 | 5 | −0.0414 [−0.0583, −0.0238] | falsified, significantly worse |
| Six-model ensemble (APTOS, external) | E10 | 5 | +0.0025 [−0.0016, +0.0066] | falsified |

## Supplementary Note 1. Decision rules

Every model emits, for DR, four logits whose sigmoids estimate $P(y > k)$, $k = 0, \ldots, 3$ (cumulative head and CORAL), or conditional probabilities $P(y > k \mid y > k-1)$ that are chained by cumulative product (CORN). From these the expected grade $s = \sum_k P(y > k)$ is computed. The five rules are:

**R1 (shipped).** Grade $= \#\{k : P(y > k) > 0.5\}$ after enforcing monotonicity by a running minimum. This is what the trained model outputs if deployed unchanged.

**R2 (development QWK cut-points).** Four ordered cut-points on $s$ chosen to maximise QWK on the development out-of-fold predictions, by coordinate ascent over a grid of 60 candidate values per cut-point (8 rounds), then applied unchanged externally.

**R3 (R2 with a referral constraint).** R2 with the grade-1/2 cut-point replaced by the largest value that keeps development referable-DR sensitivity at or above 0.90, the neighbouring cut-points pushed to preserve order.

**R4 and R4m (external, diagnostic).** The same search on the external labels, with the external set split at random into two halves (seed 0); cut-points fitted on one half are applied to the other and vice versa, so no image is graded by cut-points fitted on its own label. R4 maximises QWK, R4m macro-averaged recall. The grid has 80 candidate values.

R4/R4m use external labels and are never reported as performance: they bound what a local recalibration of the decision rule alone, without retraining, can recover. Changing the development grid from 60 to 80 values changes E08's matched development QWK by 0.0033; 60 is used for all development quantities because it reproduces the archived matched comparisons exactly.

## Supplementary Note 2. The DME grade cannot be derived from published lesion masks

IDRiD defines its DME risk grade by the distance from the fovea to the nearest hard exudate: grade 0, no exudate; grade 1, exudates more than one disc diameter from the fovea; grade 2, within one disc diameter. Because the definition is geometric, a grade derived from the pixel-level hard-exudate masks that IDRiD releases for 81 images, together with its fovea and optic-disc annotations, should reproduce the expert grade if the grade is what its definition says. The protocol was pre-registered (`docs/PARTA_preregistration.md`).

Of the 81 segmentation images, 54 could be linked to their grading-set copies by pixel correlation (minimum correlation 0.9999). The disc diameter was measured from the optic-disc mask in three ways (major axis, minor axis, equivalent-area diameter); all three give the same grades. The derived grade matched the expert grade in 96.3% of images, against 92.6% for a constant predictor, because almost every segmentation image is grade 2. On the 4 images where the test is informative (expert grade 0 or 1), the derivation failed twice: an image graded 1 whose nearest exudate lies 53 pixels from the fovea (disc diameter 612 pixels), and an image graded 0 with a single exudate 87 pixels from the fovea (disc diameter 657 pixels). In both, a single small exudate inside the threshold is either not counted by the grader or not visible at grading resolution. The pre-registered gate (exact agreement above the constant predictor on the informative images) therefore failed, and no derived DME labels were used anywhere.

## Supplementary Note 3. A per-grade verdict that depends on the cut-point procedure

The development record that motivated the matched-calibration rule included a source control (E12): two runs identical except for whether Messidor-2 images came from the default or the

native-resolution files, compared on the 953 development images both could use. Moderate recall, native minus default, under four ways of setting both arms' cut-points:

| Cut-points for both arms | Moderate recall difference (points) [95% CI] |
|---|---|
| Shipped (0.5) | −6.9 [−12.1, −2.1] |
| Fitted on all rows (in-sample) | −10.8 [−16.2, −5.4] |
| Random two-fold cross-fit, grid 80 | −6.6 [−12.0, −1.1] |
| Fold-wise cross-fit, grid 60 (development standard) | +0.4 [−5.7, +6.1] |

The originally archived value, which the project had cited as the case in which a difference *survived* matched calibration, is reproduced by the random-half procedure; under the fold-wise procedure used for every other development comparison it dissolves. Aggregate QWK is indistinguishable under every procedure. Two consequences are drawn in the main text: a matched-calibration claim must name its fitting procedure, and per-grade claims should be reported under at least two. The two-directional property of the rule (that it can confirm as well as dissolve a difference) rests instead on EfficientNet-B3 versus the 560-pixel DenseNet121, which survives matched calibration on the development pool (+0.0202 shipped, +0.0207 [+0.0105, +0.0320] matched), and on per-grade external differences that stay significant at every rule (Table 3 of the main text).

## Supplementary Note 4. Correction of an archived ordinal-head comparison

Regenerating every development number from the archived per-image logits with independent code reproduced each archived matched QWK exactly except one. The archived comparison of the CORN head against the reference reported −0.0479 [−0.0656, −0.0325], a significant deficit, with matched QWK 0.8168 for CORN. The archived script had decoded CORN's conditional logits as if they were cumulative; decoding them by cumulative product, as the training code does, gives matched QWK 0.8564 and a difference of −0.0082 [−0.0207, +0.0036], indistinguishable from the reference. The pre-registered verdict (no improvement) is unchanged; the claim that CORN is worse, and the mechanism once offered for it, are withdrawn. The script was corrected to read each run's head from its results file (repository `ISSUES.md`, §30).

## Supplementary Note 5. Seed variation (O1–O3)

The reference configuration was trained with 5 different seeds on identical data, split and recipe. Matched development QWK ranged 0.8646–0.8764 (s.d. 0.0043); the yardstick for a difference beyond seed noise between two configurations is therefore $2\sqrt{2}$ s.d. = 0.0123 (grey band in Fig. 6). On APTOS under the shipped rule, QWK ranged 0.8900–0.8968 (s.d. 0.0026). Of 50 per-grade recall differences between pairs of seeds, 15 had a bootstrap interval excluding zero under R1 (largest absolute difference 8.2 points) and 29 under R4: the pre-registered prediction O2 that seeds alone produce more than five "significant" per-grade differences held. Mild recall under the shipped rule was 5.4–6.2% and under R4m 63.8–72.7% in every seed (O3 held).

**Table S4: Seed variation of the reference configuration (O1–O3).** Five independent training seeds of E08 (identical data, split and recipe).

| Seed | Dev QWK (matched) | APTOS QWK (R1) | APTOS mild R1 (%) | APTOS mild R4m (%) | DDR QWK (R1) |
|---|---|---|---|---|---|
| E08 | 0.8646 | 0.897 | 5.9 | 70.8 | 0.733 |
| S1E08 | 0.8685 | 0.896 | 5.4 | 70.3 | 0.717 |
| S2E08 | 0.8714 | 0.895 | 6.2 | 72.7 | 0.704 |
| S3E08 | 0.8764 | 0.890 | 5.9 | 72.4 | 0.728 |
| S4E08 | 0.8692 | 0.894 | 5.9 | 63.8 | 0.713 |

## Supplementary Note 6. Reverse direction (O4–O6)

To test whether the mild-grade collapse is specific to transfer from the development pool, the reference recipe was trained on APTOS 2019 (its own frozen 5-fold split, near-duplicates grouped) and evaluated on Messidor-2 and IDRiD, the direction in which a published ordinal-regression model was reported to lose the mild grade. Matched out-of-fold QWK on APTOS was 0.9134. On Messidor-2 under the shipped rule mild recall was 34.4% (O4 was falsified); external cut-points (R4m) raised it to 44.4% without retraining (O5 was falsified). The mean predicted minus true grade under the shipped rule was −0.26 (O6, under-grading in the reverse direction, held).

**Table S5: Reverse direction (O4–O6).** The reference recipe trained on APTOS 2019 and evaluated on Messidor-2 and IDRiD.

| Corpus | Rule | QWK | Mild recall (%) | Mild F1 | Sens. (%) | Spec. (%) | Pred.−true grade |
|---|---|---|---|---|---|---|---|
| Messidor-2 | R1 | 0.715 | 34.4 | 0.276 | 43.5 | 99.2 | −0.26 |
| | R2 | 0.705 | 16.3 | 0.168 | 47.3 | 98.9 | −0.32 |
| | R3 | 0.679 | 17.0 | 0.155 | 34.4 | 99.7 | −0.36 |
| | R4 | 0.779 | 25.9 | 0.273 | 77.9 | 93.8 | −0.02 |
| | R4m | 0.755 | 44.4 | 0.325 | 82.7 | 91.7 | +0.16 |
| IDRiD | R1 | 0.807 | 64.0 | 0.239 | 81.7 | 96.9 | −0.24 |
| | R2 | 0.827 | 36.0 | 0.200 | 85.4 | 95.3 | −0.23 |
| | R3 | 0.810 | 44.0 | 0.173 | 75.5 | 97.9 | −0.31 |
| | R4 | 0.841 | 28.0 | 0.200 | 88.2 | 92.7 | −0.04 |
| | R4m | 0.828 | 56.0 | 0.231 | 88.9 | 92.2 | +0.08 |

## Supplementary Note 7. Content-overlap audit

Public fundus corpora are assembled from overlapping sources, so a copy of an external image in the training or pretraining data would inflate external performance. Every image of the development pool, APTOS, DDR and EyePACS was reduced to a 256-bit difference hash after the same retinal crop used for training. In this project's data, re-encodings of one photograph lie within 8 bits of each other and distinct photographs at least 28 bits apart, with a grey zone between; all pairs within 16 bits were counted and every pair within 8 bits was archived.

*Development pool and external sets.* No development image lay within 8 bits of any APTOS or DDR image (closest pairs 12 and 14 bits), so no external estimate is affected by development-set leakage.

*EyePACS (pretraining) and external sets.* No external image was hash-identical to an EyePACS image. 18 APTOS–EyePACS pairs (11 APTOS images) and 2,304 DDR–EyePACS pairs (202 DDR images) lay within 8 bits, the closest at 6 and 3 bits. Their distances form a tail that rises towards the threshold rather than a peak at zero, the pattern expected of hash collisions between dark, low-contrast photographs rather than of copies; we could not verify them visually. Removing every flagged image changed no model's shipped-rule referable sensitivity, mild recall, QWK or referral area under the curve by more than 0.55 percentage points on DDR and 0.04 on APTOS.

*Within corpora.* The development pool contains 10 hash-identical pairs, 7 of which straddle two folds of the frozen split (14 images, missed when the split was built); removing those images changes every model's matched development QWK by at most 0.0008. APTOS and DDR contain 148 and 120 hash-identical pairs respectively. Within-corpus duplicates in an external set do not leak information into the model, but they make image-level bootstrap intervals slightly narrow.

**Table S6: Content-overlap audit.** Pairs of images across corpora whose 256-bit difference hashes lie within the stated Hamming distance.

| Corpus pair | Images | Pairs ≤16 bits | ≤8 bits | 0 bits |
|---|---|---|---|---|
| within Development pool | 2,260 | 29 | 10 | 10 |
| Development pool vs APTOS | 2,260 / 3,662 | 47 | 0 | 0 |
| within APTOS | 3,662 | 1,355 | 149 | 148 |
| Development pool vs DDR | 2,260 / 12,522 | 14 | 0 | 0 |
| within DDR | 12,522 | 23,021 | 265 | 120 |
| APTOS vs DDR | 3,662 / 12,522 | 1,361 | 1 | 0 |
| EyePACS vs APTOS | 35,126 / 3,662 | 10,086 | 18 | 0 |
| EyePACS vs DDR | 35,126 / 12,522 | 267,402 | 2,304 | 0 |
| EyePACS vs Development pool | 35,126 / 2,260 | 178 | 0 | 0 |
| Threshold 16 bits; development pairs split across folds: 7 | | | | |

## Supplementary Tables S7–S8. Full decision-rule matrices

**Table S7: Full decision-rule matrix, APTOS.** QWK, accuracy, per-grade recall (%) and referable-DR sensitivity/specificity (%) of every model under every rule. R4/R4m are fitted on the external labels by two-fold cross-fitting and are diagnostic bounds, not performance.

| Model | Rule | QWK | Acc. | No DR | Mild | Mod. | Sev. | PDR | Sens. | Spec. |
|---|---|---|---|---|---|---|---|---|---|---|
| E08 | R1 | 0.897 | 73.5 | 97.0 | 5.9 | 64.2 | 64.2 | 52.2 | 99.5 | 84.3 |
| | R2 | 0.883 | 68.2 | 98.3 | 8.9 | 39.0 | 56.0 | 65.4 | 99.1 | 85.1 |
| | R3 | 0.882 | 67.9 | 98.3 | 4.6 | 39.2 | 56.0 | 65.4 | 99.3 | 84.2 |
| | R4 | 0.904 | 78.2 | 98.8 | 18.1 | 77.9 | 49.7 | 47.8 | 96.6 | 87.0 |
| | R4m | 0.886 | 75.2 | 96.6 | 70.8 | 49.0 | 47.2 | 56.3 | 80.4 | 95.4 |
| E09 | R1 | 0.879 | 69.3 | 88.5 | 4.3 | 66.8 | 62.7 | 45.8 | 99.3 | 84.1 |
| | R2 | 0.863 | 62.0 | 88.5 | 22.7 | 30.3 | 62.2 | 56.6 | 96.3 | 87.4 |
| | R3 | 0.847 | 61.0 | 88.5 | 1.4 | 34.3 | 62.2 | 56.6 | 99.7 | 79.7 |
| | R4 | 0.898 | 76.7 | 98.6 | 10.0 | 75.5 | 48.7 | 48.5 | 98.6 | 85.9 |
| | R4m | 0.886 | 75.6 | 97.8 | 61.6 | 53.6 | 50.3 | 48.5 | 81.1 | 94.7 |
| E10 | R1 | 0.887 | 70.8 | 94.4 | 4.3 | 59.9 | 72.5 | 45.8 | 99.0 | 84.2 |
| | R2 | 0.885 | 70.1 | 98.1 | 7.0 | 50.3 | 42.0 | 63.1 | 98.8 | 84.7 |
| | R3 | 0.884 | 69.8 | 98.1 | 3.2 | 50.8 | 42.0 | 63.1 | 99.2 | 84.0 |
| | R4 | 0.898 | 76.8 | 98.0 | 13.5 | 75.5 | 50.3 | 49.2 | 97.2 | 85.9 |
| | R4m | 0.876 | 73.0 | 97.8 | 77.8 | 36.6 | 50.8 | 52.5 | 72.6 | 97.0 |
| E11FULL | R1 | 0.903 | 74.9 | 98.1 | 4.9 | 65.6 | 63.2 | 60.3 | 99.3 | 84.3 |
| | R2 | 0.900 | 72.9 | 98.9 | 7.6 | 54.5 | 63.2 | 64.1 | 99.0 | 84.9 |
| | R3 | 0.900 | 72.8 | 98.9 | 7.0 | 54.6 | 63.2 | 64.1 | 99.1 | 84.8 |
| | R4 | 0.907 | 79.1 | 98.6 | 16.8 | 81.8 | 39.4 | 54.6 | 96.5 | 86.6 |
| | R4m | 0.890 | 73.4 | 97.6 | 75.1 | 37.3 | 51.3 | 59.7 | 75.3 | 96.4 |
| E14MAC | R1 | 0.880 | 68.7 | 95.5 | 5.7 | 53.9 | 72.5 | 31.9 | 98.7 | 84.3 |
| | R2 | 0.879 | 67.2 | 98.6 | 5.7 | 39.6 | 72.0 | 42.7 | 98.3 | 84.7 |
| | R3 | 0.877 | 67.1 | 98.6 | 1.4 | 40.9 | 72.0 | 42.7 | 99.2 | 83.6 |
| | R4 | 0.889 | 74.6 | 98.9 | 10.3 | 69.0 | 52.8 | 40.3 | 97.6 | 85.7 |
| | R4m | 0.857 | 71.5 | 97.4 | 75.7 | 35.7 | 44.0 | 46.8 | 71.2 | 96.6 |
| E15LPFT | R1 | 0.874 | 65.3 | 89.6 | 5.1 | 45.1 | 70.5 | 57.6 | 99.2 | 84.2 |
| | R2 | 0.880 | 69.2 | 97.2 | 10.3 | 43.5 | 57.0 | 66.1 | 98.3 | 85.3 |
| | R3 | 0.878 | 68.9 | 97.2 | 5.1 | 44.5 | 57.0 | 66.1 | 99.1 | 84.2 |
| | R4 | 0.898 | 77.3 | 98.6 | 11.9 | 77.5 | 44.6 | 49.2 | 97.3 | 86.0 |
| | R4m | 0.878 | 75.5 | 97.7 | 70.3 | 49.3 | 34.2 | 61.4 | 78.5 | 96.0 |
| E17NAT | R1 | 0.891 | 71.1 | 96.1 | 5.9 | 56.8 | 72.0 | 47.8 | 99.2 | 84.6 |
| | R2 | 0.872 | 65.5 | 97.3 | 9.5 | 32.2 | 48.2 | 64.7 | 99.0 | 85.0 |
| | R3 | 0.870 | 64.9 | 97.3 | 3.2 | 32.6 | 48.2 | 64.7 | 99.4 | 83.7 |
| | R4 | 0.894 | 76.5 | 99.1 | 16.2 | 72.0 | 48.7 | 47.8 | 96.3 | 86.9 |
| | R4m | 0.874 | 73.6 | 97.7 | 72.7 | 41.7 | 50.3 | 51.2 | 74.0 | 96.3 |
| E20CORAL | R1 | 0.879 | 66.3 | 96.1 | 14.3 | 39.1 | 64.8 | 42.4 | 98.2 | 85.3 |
| | R2 | 0.869 | 65.7 | 98.1 | 13.8 | 33.6 | 47.7 | 53.2 | 98.2 | 85.4 |
| | R3 | 0.867 | 65.2 | 98.1 | 4.1 | 35.2 | 47.7 | 53.2 | 99.4 | 83.3 |
| | R4 | 0.887 | 75.2 | 97.8 | 15.7 | 73.8 | 35.8 | 42.4 | 97.1 | 85.7 |
| | R4m | 0.872 | 73.1 | 97.3 | 72.4 | 45.4 | 48.2 | 35.6 | 79.4 | 95.5 |
| E22CORN | R1 | 0.892 | 72.7 | 97.6 | 1.6 | 62.3 | 71.0 | 45.8 | 99.3 | 83.9 |
| | R2 | 0.889 | 69.3 | 97.0 | 10.8 | 47.3 | 76.2 | 43.1 | 98.4 | 85.3 |
| | R3 | 0.887 | 69.1 | 97.0 | 5.4 | 48.5 | 76.2 | 43.1 | 99.3 | 84.0 |
| | R4 | 0.896 | 77.2 | 98.2 | 14.6 | 76.3 | 54.4 | 44.7 | 96.9 | 86.4 |
| | R4m | 0.875 | 75.0 | 96.8 | 77.0 | 48.9 | 43.0 | 47.8 | 74.1 | 96.7 |
| E21CNXA | R1 | 0.862 | 62.9 | 84.7 | 8.9 | 48.5 | 69.4 | 41.7 | 98.7 | 84.5 |
| | R2 | 0.860 | 62.0 | 96.0 | 13.5 | 21.4 | 62.7 | 52.2 | 97.5 | 85.8 |
| | R3 | 0.855 | 61.3 | 96.0 | 1.4 | 23.4 | 62.7 | 52.2 | 99.2 | 82.3 |
| | R4 | 0.890 | 76.1 | 97.8 | 18.6 | 74.4 | 51.3 | 37.3 | 96.8 | 87.1 |
| | R4m | 0.873 | 74.3 | 98.0 | 67.0 | 48.3 | 51.8 | 41.0 | 79.6 | 95.3 |

**Table S8: Full decision-rule matrix, DDR.** QWK, accuracy, per-grade recall (%) and referable-DR sensitivity/specificity (%) of every model under every rule. R4/R4m are fitted on the external labels by two-fold cross-fitting and are diagnostic bounds, not performance.

| Model | Rule | QWK | Acc. | No DR | Mild | Mod. | Sev. | PDR | Sens. | Spec. |
|---|---|---|---|---|---|---|---|---|---|---|
| E08 | R1 | 0.733 | 73.7 | 97.7 | 33.3 | 51.2 | 58.5 | 50.3 | 65.5 | 98.7 |
| | R2 | 0.734 | 68.9 | 98.8 | 22.1 | 38.2 | 53.4 | 50.3 | 62.5 | 99.0 |
| | R3 | 0.742 | 70.5 | 98.8 | 17.5 | 43.4 | 53.4 | 50.3 | 67.1 | 98.3 |
| | R4 | 0.805 | 69.8 | 86.2 | 25.2 | 57.8 | 57.2 | 49.2 | 80.8 | 93.2 |
| | R4m | 0.787 | 67.0 | 83.6 | 59.8 | 49.8 | 55.9 | 45.3 | 66.8 | 98.4 |
| E09 | R1 | 0.697 | 68.8 | 93.5 | 20.5 | 46.4 | 54.7 | 46.3 | 63.9 | 97.3 |
| | R2 | 0.708 | 58.6 | 92.0 | 31.7 | 18.1 | 56.8 | 47.0 | 50.4 | 99.2 |
| | R3 | 0.733 | 66.5 | 92.0 | 11.7 | 43.0 | 56.8 | 47.0 | 72.1 | 93.4 |
| | R4 | 0.753 | 60.7 | 77.1 | 38.4 | 43.6 | 48.3 | 50.4 | 70.9 | 94.1 |
| | R4m | 0.744 | 57.3 | 71.6 | 51.6 | 41.2 | 61.0 | 41.4 | 64.9 | 96.9 |
| E10 | R1 | 0.737 | 70.5 | 96.1 | 43.7 | 44.8 | 59.3 | 42.7 | 62.8 | 98.2 |
| | R2 | 0.740 | 69.3 | 97.4 | 27.1 | 41.9 | 41.9 | 47.8 | 63.4 | 98.2 |
| | R3 | 0.748 | 71.1 | 97.4 | 21.6 | 47.6 | 41.9 | 47.8 | 68.3 | 97.4 |
| | R4 | 0.793 | 67.2 | 85.1 | 23.5 | 53.1 | 67.4 | 43.3 | 80.4 | 92.6 |
| | R4m | 0.775 | 63.4 | 83.0 | 57.9 | 40.9 | 58.5 | 43.4 | 66.7 | 97.6 |
| E11FULL | R1 | 0.755 | 76.1 | 98.2 | 29.8 | 56.7 | 55.5 | 56.1 | 70.3 | 98.7 |
| | R2 | 0.756 | 72.5 | 98.9 | 22.1 | 46.9 | 56.4 | 55.6 | 65.2 | 99.0 |
| | R3 | 0.757 | 72.6 | 98.9 | 21.4 | 47.4 | 56.4 | 55.6 | 65.6 | 98.9 |
| | R4 | 0.813 | 72.8 | 89.2 | 24.6 | 60.2 | 60.6 | 58.4 | 81.2 | 94.2 |
| | R4m | 0.799 | 66.7 | 84.2 | 58.7 | 46.3 | 64.8 | 53.1 | 69.4 | 98.4 |
| E14MAC | R1 | 0.736 | 70.2 | 96.3 | 42.1 | 43.7 | 71.2 | 39.6 | 61.7 | 98.9 |
| | R2 | 0.729 | 68.1 | 98.6 | 19.5 | 37.1 | 62.7 | 45.2 | 62.5 | 98.8 |
| | R3 | 0.740 | 70.4 | 98.6 | 10.2 | 44.9 | 62.7 | 45.2 | 69.4 | 97.4 |
| | R4 | 0.793 | 67.8 | 84.7 | 24.0 | 55.2 | 61.9 | 45.5 | 80.1 | 93.1 |
| | R4m | 0.776 | 64.8 | 84.4 | 57.1 | 43.8 | 68.2 | 38.4 | 65.6 | 98.3 |
| E15LPFT | R1 | 0.764 | 69.9 | 92.5 | 45.2 | 43.7 | 63.1 | 62.3 | 66.1 | 98.3 |
| | R2 | 0.743 | 68.9 | 97.5 | 21.4 | 38.2 | 51.7 | 60.0 | 62.2 | 98.8 |
| | R3 | 0.753 | 71.2 | 97.5 | 16.5 | 45.4 | 51.7 | 60.0 | 68.4 | 97.9 |
| | R4 | 0.794 | 67.5 | 84.6 | 35.2 | 52.0 | 66.5 | 48.6 | 76.8 | 94.5 |
| | R4m | 0.786 | 64.4 | 81.0 | 57.6 | 44.5 | 61.0 | 54.0 | 67.6 | 97.9 |
| E17NAT | R1 | 0.743 | 71.5 | 96.9 | 38.9 | 46.4 | 58.9 | 45.5 | 64.6 | 98.2 |
| | R2 | 0.765 | 66.9 | 96.1 | 33.8 | 35.0 | 54.7 | 48.5 | 65.2 | 98.3 |
| | R3 | 0.772 | 68.9 | 96.1 | 26.5 | 41.6 | 54.7 | 48.5 | 71.0 | 97.0 |
| | R4 | 0.801 | 67.5 | 85.7 | 31.4 | 51.8 | 55.5 | 48.2 | 79.4 | 93.4 |
| | R4m | 0.789 | 63.4 | 83.3 | 62.1 | 40.4 | 63.6 | 41.1 | 67.5 | 97.8 |
| E20CORAL | R1 | 0.740 | 65.2 | 93.8 | 37.8 | 36.0 | 59.3 | 33.1 | 58.9 | 98.5 |
| | R2 | 0.728 | 65.6 | 96.2 | 24.9 | 33.8 | 50.8 | 43.8 | 58.9 | 98.5 |
| | R3 | 0.743 | 69.6 | 96.2 | 14.4 | 46.5 | 50.8 | 43.8 | 69.7 | 96.6 |
| | R4 | 0.778 | 64.2 | 82.9 | 37.1 | 46.4 | 57.6 | 43.5 | 75.1 | 94.1 |
| | R4m | 0.768 | 60.6 | 81.0 | 58.7 | 37.1 | 64.4 | 35.5 | 65.6 | 97.7 |
| E22CORN | R1 | 0.699 | 72.4 | 98.7 | 25.9 | 49.1 | 56.4 | 42.5 | 63.8 | 98.6 |
| | R2 | 0.707 | 66.5 | 98.0 | 32.4 | 34.7 | 64.0 | 30.0 | 54.7 | 99.3 |
| | R3 | 0.730 | 70.2 | 98.0 | 24.0 | 46.3 | 64.0 | 30.0 | 65.0 | 98.3 |
| | R4 | 0.785 | 67.4 | 85.2 | 22.7 | 54.5 | 56.4 | 41.9 | 79.4 | 92.8 |
| | R4m | 0.766 | 63.6 | 85.9 | 56.5 | 39.1 | 63.1 | 36.3 | 63.2 | 98.6 |
| E21CNXA | R1 | 0.710 | 65.4 | 88.0 | 34.9 | 45.0 | 51.7 | 34.7 | 62.6 | 98.4 |
| | R2 | 0.701 | 62.1 | 94.9 | 23.8 | 26.3 | 53.8 | 40.4 | 55.1 | 99.2 |
| | R3 | 0.726 | 67.8 | 94.9 | 8.7 | 44.5 | 53.8 | 40.4 | 71.3 | 95.8 |
| | R4 | 0.747 | 58.7 | 78.5 | 41.4 | 37.3 | 56.4 | 40.3 | 70.0 | 96.3 |
| | R4m | 0.738 | 56.9 | 70.5 | 53.5 | 43.0 | 59.3 | 32.5 | 67.4 | 97.3 |